THE HNS ORIGIN TRILOGY

Brain Architecture as the Foundation of Trustworthy AI

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github.com/satoru-hara/03_NSW

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Preface

The Reliability Crisis in Artificial Intelligence

Artificial intelligence is at a turning point. Large language models now generate text indistinguishable from human writing, assist in scientific research, advise on medical and legal questions, and participate in decisions affecting millions of lives. Yet they fail in ways that human experts do not. They assert falsehoods with confidence. They lose track of the question being asked. They draw causal conclusions from mere correlations. They let a metaphor quietly replace a fact. They drift, without signalling, from the topic at hand to something adjacent but different.

These failures are not bugs in the implementation. They are structural properties of the architecture. A language model generates its output by sampling, at every step, from a high-dimensional probability distribution over the vocabulary. The output is fluent because high-probability sequences are linguistically coherent. But fluency and validity are independent properties. A sentence can be statistically natural and causally groundless. It can be grammatically perfect and factually empty. Probability does not confer structure.

The dominant response to this problem has been to scale: larger models, trained on larger datasets, subject to more reinforcement learning from human feedback. This approach has produced impressive capabilities. It has not solved the structural problem. A larger model makes fewer surface errors, but it remains susceptible to the same classes of deep failure — Layer Jump, Category Ambiguity, Scope Drift, Metaphor Contamination, Unsupported Causality — because those failures arise from a property of the architecture that size does not change. The model operates on a single axis of computation, with no independent mechanism to verify whether its outputs are structurally valid, causally grounded, or contextually coherent.

The question this book asks is: where does reliable cognition come from? Not in AI systems, but in the system we know produces it — the human brain. If we can understand the mechanism by which the brain achieves reliability despite operating on a stochastic substrate, we may be able to engineer that mechanism into artificial systems. This is the programme of Human Natural Structure (HNS), and The HNS Origin Trilogy is its foundational documentation.

About This Book

This book collects the three foundational papers of the HNS research programme, written as a unified trilogy in which each paper addresses a distinct level of the same problem. The papers are presented in logical order: from the neuroscientific theory that motivates the programme (Paper I), through the engineering architecture that implements it (Paper II), to the structural correspondence that validates it (Paper III). Each chapter is self-contained but gains depth from the others; the trilogy is designed to be read as a whole.

The intended reader is anyone who takes seriously the question of whether AI systems can be made reliably trustworthy — not merely more capable. This includes researchers and engineers working on AI safety and alignment; policy makers and standards developers grappling with the challenge of making AI systems explainable, verifiable, and governable; and cognitive scientists and neuroscientists interested in the implications of the MAV framework for understanding disorders of human cognition. No prior knowledge of information geometry or neural network architecture is assumed.

Structure of the Trilogy

No.	Title	Domain	Core Contribution
I	The Multi-Axis Verification Process of the Human Brain	Computational Neuroscience	Formalises MAV as the brain's mechanism for reliable cognition from stochastic components
II	Human Natural Structure: Implementing the Brain's MAV for Advanced AI Alignment	AI Safety & Alignment	Engineers MAV into LLM pipelines via HNS-36/144/864 and SMS-6; achieves zero structural hallucination
III	Human Brain and AI with HNS: A Correspondence Table	Cognitive Systems Engineering	Establishes the structural isomorphism between biological cognition and HNS-aligned AI

Table P.1. The three papers of The HNS Origin Trilogy.

The complete HNS-PoC-Package v1.0, including the hns36 runtime implementation, the HNS Structural Hallucination Taxonomy, and the External Verification Architecture (EVA) specification, is available at: github.com/satoru-hara/03_NSW

Chapter I

The Multi-Axis Verification Process of the Human Brain

A Brain-Inspired Architecture for Geometric Error-Correction in Stochastic Cognitive Systems

1.1 The Reliability Paradox

Consider a pyramidal neuron in the prefrontal cortex. At any given moment, it integrates thousands of excitatory and inhibitory synaptic inputs, each itself stochastic. The probability that a synaptic vesicle releases its neurotransmitter on any given action potential typically falls between 0.1 and 0.9. Whether the neuron itself fires is probabilistic in a way that cannot be reduced away by averaging. And yet: the trained surgeon's scalpel moves to within a fraction of a millimetre of its target. The mathematician maintains the thread of a proof across twenty minutes of working memory. The native speaker resolves the meaning of an ambiguous sentence in under 200 milliseconds.

This is the reliability paradox of neuroscience: how does a system composed entirely of noisy, unreliable components produce outputs of such remarkable fidelity? The standard answer — that population coding averages away the noise of individual neurons — is correct but insufficient. Averaging N independent neurons reduces variance by a factor of 1/N, but the brain contains approximately 86 billion neurons and still makes systematic errors: confabulation, hallucination, categorical confusion, delusional causal reasoning. These errors are not residues of insufficient averaging. They are structurally patterned, occur in specific situations, and map onto recognisable diagnostic categories. A complementary mechanism must be at work.

1.2 The MAV Hypothesis

The Multi-Axis Verification Process (MAV) proposes that the brain's primary error-suppression mechanism is not the averaging of noise within a single processing axis, but the geometric intersection of three mutually orthogonal verification processes operating simultaneously on different cognitive axes. The core claim: the human brain routinely instantiates three independent constraint structures — the Stochastic/Generative Axis, the Structural/Causal Axis, and the Grounded/Executive Axis — and uses their simultaneous intersection to identify the unique cognitive output that satisfies all three constraint classes simultaneously.

This output is called $X^{\mathrm{In}}{}_{\mathrm{v}}{}^{\mathrm{argant}}$: the invariant point of the triple intersection. It is invariant in the sense that small perturbations along any single axis do not displace it from the feasible region, because the other two axes re-anchor it. The MAV hypothesis has a specific mathematical structure. When three constraint surfaces are mutually orthogonal in the information-geometric sense — when the Fisher information matrix induces zero correlation between the parameters defining each surface — their intersection does not merely add their individual error reductions; it multiplies them:

> $Vol(A_1 \cap A_2 \cap A_3) \approx Vol(A_1) \cdot Vol(A_2) \cdot Vol(A_3) / Vol(\mathcal{M})^2$

This product is orders of magnitude smaller than any individual axis volume. The geometric intersection achieves error reduction that no single axis, however powerful, can achieve alone. This is a consequence of the information geometry of orthogonal manifold intersection — not a metaphor, but a mathematical result.

1.3 The Three Axes

Axis 1: Stochastic / Generative Axis

Implemented by cortical association areas — the default mode network, lateral temporal cortex, and parietal association cortex — Axis 1 is the generative engine of cognition. It integrates statistical regularities abstracted from experience and generates rapidly-available, high-prior-probability response candidates within tens of milliseconds. Formally: $A_1 = \{x \in \mathcal{M} \mid P(x \mid \text{context}, \theta) \geq \varepsilon_1\}$. The volume of A_1 is large: many valid and many invalid outputs share high prior probability. Axis 1 provides generativity and speed; it does not provide validity.

Axis 2: Structural / Causal Axis

Implemented by the prefrontal cortex–hippocampal network, Axis 2 defines the permissible structure of cognitive outputs, independent of statistical frequency. The dorsolateral PFC maintains hierarchical representations of causal dependencies; the hippocampus binds episodic sequences into temporally-ordered schemas and enforces category boundaries across contextual shifts. Together, these systems enforce: (a) category boundary conditions

— preventing the conflation of cognitive, emotional, and intentional representations; (b) causal ordering constraints — requiring causal claims to be grounded in the current event schema; and (c) hierarchical scope conditions — requiring explicit transitions between abstraction levels. Formally: $A_2 = \{x \in \mathcal{M} \mid \forall i: c_i(x) = \text{TRUE}\}$.

Axis 3: Grounded / Executive Axis

Implemented by the sensorimotor system, the anterior cingulate cortex (ACC), and the predictive coding hierarchy of the posterior parietal and premotor cortices, Axis 3 constitutes the brain's 'reality friction interface.' The ACC monitors conflict between predicted and actual outcomes, signalling when the output trajectory is inconsistent with environmental reality. An output may be both statistically plausible and structurally valid but still be contextually misaligned — a valid response to the wrong question. Axis 3 detects and suppresses such misalignment. Formally: $A_3 = \{x \in \mathcal{M} \mid G(x) = 1\}$, where G is a dynamic grounding function continuously updated in real time.

1.4 The Orthogonality Conditions

The error-cancellation power of the MAV depends on the mutual orthogonality of the three axes. Three independent arguments support the approximate satisfaction of this condition:

- Axis 1 ⊥ Axis 2: Statistical co-occurrence frequency is independent of causal validity. Causally invalid propositions can be statistically common; causally valid ones can be statistically rare. The parameters of θ are logically independent of the causal constraint set C.
- Axis 2 ⊥ Axis 3: Structural constraints are defined over the abstract category and causal space; grounding constraints are defined over the current environmental state. A proposition can be structurally valid but contextually irrelevant, or contextually grounded but structurally misframed.
- Axis 1 ⊥ Axis 3: Prior distributions are defined over the training-time distribution, which can differ substantially from the current context. This independence is most evident in novel or unusual contexts, where high-probability prior outputs may be contextually invalid.

Perfect orthogonality is an idealisation; real neural systems achieve approximate orthogonality. The residual correlation between axes defines the system's irreducible error floor — the class of errors that persist even when all three axes are active, likely corresponding to the systematic cognitive biases universal across individuals and cultures.

1.5 Failure Modes and Clinical Correlates

The MAV framework generates a specific prediction: selective disruption of individual axes will produce characteristically distinct failure modes. This prediction is directly testable against the phenomenology of neurological and psychiatric conditions.

Failure Type	Axis	Definition	HNS Suppression	Clinical Correlate
Layer Jump	Axis 2	Traversal of meaning layers without explicit bridging	HNS-36 + SMS-6	Formal thought disorder (schizophrenia)
Category Ambiguity	Axis 2	Cognitive, emotional & intentional categories conflated	HNS-144 + SMS-5	Semantic dementia; acute psychosis
Scope Drift	Axis 3	Departure from intended scope without signalling	SMS-2 + SMS-3	Confabulation in frontal lobe syndrome
Metaphor Contamination	Axis 3	Figurative expressions displace structural coordinates	SMS-1 + SMS-6	Metaphorical concretisation in schizotypy
Unsupported Causality	Axes 2+3	Causal claim without mechanistic grounding	HNS-864 + SMS-4	Delusional reasoning; confabulatory amnesia

Table 1.1. MAV axis failure modes, HNS suppression layers, and clinical correlates.

This mapping is predictive, not merely descriptive. Conditions primarily affecting the PFC–hippocampal network (Axis 2) should show disproportionately high rates of Layer Jump and Category Ambiguity, with relative preservation of contextual grounding. Conditions primarily affecting the sensorimotor–ACC system (Axis 3) should show disproportionate Scope Drift and Metaphor Contamination. Joint Axis 2 and 3 failure should produce the highest rates of Unsupported Causality. These predictions are amenable to quantitative testing with structured cognitive assessments paired with neuroimaging measures of axis-specific activity.

1.6 Implications for Artificial Intelligence

The MAV framework has a direct implication for AI systems: any system that operates on a single computational axis — as current LLMs do — is structurally incapable of achieving the reliability that the MAV architecture provides. No degree of scaling on Axis 1 can substitute for the independent constraint structures of Axes 2 and 3. The error volume achievable by a single-axis strategy has a hard lower bound that is independent of the tightness of that axis' constraint. This conclusion is the foundation for the engineering programme presented in Chapter II.

Chapter II

Human Natural Structure

Implementing the Brain's Multi-Axis Verification Process for Advanced AI Alignment

2.1 The Structural Inadequacy of Current Alignment

The dominant AI alignment paradigms — Reinforcement Learning from Human Feedback (RLHF), prompt-based guardrails, and Retrieval-Augmented Generation (RAG) — share a common structural limitation: they are single-axis interventions. They modify, constrain, or supplement the Stochastic/Generative Axis without activating an independent, orthogonal constraint structure. In geometric terms, they tighten the feasible region on a single hypersurface A_1 without intersecting it with A_2 and A_3. The residual error volume remains high, and all five structural hallucination types remain structurally possible.

RLHF modifies the model's weight distribution by rewarding preferred outputs, but alignment is encoded in the same probabilistic substrate that generates errors; the error space is reshaped, not removed. Prompt-based guardrails intercept outputs after generation and block surface-level threat patterns, but cannot address structural incoherence that presents as plausible natural language. RAG supplements the generative process with retrieved facts, but without a structural constraint layer, retrieved content can be causally misapplied or categorically misframed.

What current AI alignment architectures lack is the engineering instantiation of Axes 2 and 3 as genuinely independent, orthogonal constraint planes that intersect the LLM's generative output in real time. Human Natural Structure (HNS) addresses precisely this gap.

2.2 The HNS Architecture

Human Natural Structure is a structural operating system that instantiates the abstract MAV constraint planes within the inference pipeline of a deployed LLM. It introduces no modifications to the base model's weights or architecture; instead, it operates as an intercepting verification layer between the model's internal probability state and its decoded output. The architecture comprises two independent components:

- Axis 2 implementation: the HNS Cellular Matrix (HNS-36 / HNS-144 / HNS-864)
- Axis 3 implementation: the HNS Social Meta Structure Specification (HNS-SMS-6)

These are integrated by the hns36 runtime engine, a pre-decode interception layer that applies both constraint checks before the model's probability distribution is decoded into output tokens. The geometric intersection $A_1 \cap A_2 \cap A_3$ is computed at every token generation step.

2.3 Axis 2: The HNS Cellular Matrix

HNS-36: Base Coordinate System

The base layer is a 36-cell coordinate system organised as a 6 × 6 grid. The six rows represent the natural causal layers of event representation, spanning from physical/material substrate to macro-social systems. The six columns represent the abstract cognitive categories through which any event can be parsed: entity, process, relation, state, transformation, and meta-level. Together, these 36 cells constitute a complete tiling of the structural space within which any well-formed cognitive output must be locatable.

The primary function of HNS-36 is macro-level screening for Layer Jump: if the probability-weighted trajectory of an LLM output crosses non-adjacent cell boundaries without an explicit transitional proposition, the traversal is flagged as a structural constraint violation ($c_i(x)$ = FALSE), and the token probability of the violating continuation is attenuated.

HNS-144: Logical Relation Extension

Each of the 36 core cells is expanded along four logical relation dimensions — subjective/objective, static/dynamic, local/global, and necessary/contingent — yielding a 144-cell specification. This layer provides the granularity required for Category Ambiguity suppression: the four-dimensional logical relation structure enforces precise concept boundary conditions, preventing the conflation of dynamic processes with static entities, or subjective states with objective properties.

HNS-864: Precision Causal Audit

Each HNS-144 cell is further subdivided by six verification invariants corresponding to the six logical modalities of propositional validity, yielding an 864-cell precision audit structure. HNS-864 operates at the atomic proposition level, auditing individual causal claims for mechanistic grounding. A causal claim token can only pass the HNS-864 filter if its causal chain is traceable to a valid, contiguous cell-transition path within the constraint matrix, structurally precluding Unsupported Causality.

2.4 Axis 3: The HNS-SMS-6 Protocol

The HNS-SMS-6 (Social Meta Structure Specification) protocol is a six-layer dynamic grounding specification that tests output candidates against the actual operational context. G(x) = 1 only if the output is compatible with all six layers; G(x) = 0 triggers immediate context-coherence correction. The six layers and their grounding functions are summarised below.

Layer	Domain	Grounding Function	Failure Mode Addressed	Brain Analogue
SMS-1	Somatic	Physical & ergonomic coherence with operational environment	Metaphor Contamination	Sensorimotor feedback system
SMS-2	Protocol	Communicative alignment; mutual intelligibility	Scope Drift	ACC conflict monitoring
SMS-3	Ecosystem	Organisational & collaborative context scope	Scope Drift	Posterior parietal context tracking
SMS-4	Economic	Resource & information equivalence; value-relevance	Unsupported Causality	Orbitofrontal value evaluation
SMS-5	Governance	Conformance with external standards & normative	Category Ambiguity	PFC rule-based constraint enforcement

Layer	Domain	Grounding Function	Failure Mode Addressed	Brain Analogue
		constraints		
SMS-6	Universal	Global optimality; no higher-order systemic violation	Layer Jump	Prefrontal hierarchical coherence monitoring

Table 2.1. The six layers of the HNS-SMS-6 protocol, their grounding functions, and brain analogues.

G(x) is a time-varying function. As the conversation context evolves, the SMS-6 layers are re-evaluated at each token generation step, updating the constraint surface A_3 in real time. This dynamic character is the structural analogue of the brain's sensorimotor-ACC feedback loop, which continuously re-anchors cognitive output to the evolving environmental state.

2.5 The hns36 Runtime Engine

The engineering integration of Axes 2 and 3 with the LLM's generative pipeline (Axis 1) is implemented by the hns36 runtime engine. The runtime operates as an interception layer between the model's hidden-state computation and its final token decoding step. Token candidates whose probability vectors fail the HNS-36/144/864 constraint filters — for which any $c_i(x)$ = FALSE — are attenuated at the pre-decode stage. Simultaneously, candidates for which G(x) = 0 under SMS-6 evaluation are blocked. Only the probability mass residing in the triple intersection $A_1 \cap A_2 \cap A_3$ survives to be decoded as the final output token.

Each constraint check is logged with its specific HNS matrix coordinate (the HNS-36/144/864 coordinate triple) or SMS-6 layer reference, constituting the auditable record of the verification decision. This logging is not an optional feature; it is a structural property of the interception architecture. Every correction is attributable by construction.

2.6 Empirical Validation

The hns36 runtime was integrated with a 70-billion-parameter open-source base LLM. No fine-tuning or weight modification was applied. Three configurations were compared: baseline LLM, guardrail-augmented (rule-based filter, 512-token context window), and HNS-augmented (full hns36 runtime). Each configuration was evaluated on 50-turn high-complexity interaction sequences designed to elicit all five hallucination types, with three sequences per configuration (150 turns total), annotated by two independent reviewers.

Failure Type	Baseline LLM	Guardrail-augmented	HNS-augmented
Layer Jump	18.2%	12.4%	**0.0%**
Category Ambiguity	14.7%	11.8%	**0.0%**
Scope Drift	21.5%	15.3%	**0.0%**
Metaphor Contamination	9.3%	8.6%	**0.0%**
Unsupported Causality	16.8%	13.1%	**0.0%**
Overall (any type)	43.1%	28.9%	**0.0%**

Table 2.2. Hallucination incidence rates per 50-turn sequence by configuration and failure type.

The HNS-augmented configuration achieved zero incidence across all five failure types. The guardrail-augmented configuration reduced overall incidence from 43.1% to 28.9%, but failed to achieve zero incidence on any failure type, with residual rates of 8.6–15.3% across categories. Token generation latency for HNS-augmented was 47.8 ms (1.0% overhead vs. 47.3 ms baseline), substantially lower than guardrail-augmented (49.1 ms, +3.8%). This reflects the geometric efficiency of the HNS constraint checks: low-dimensional projection operations on the pre-decode probability vector, not full-sequence re-evaluations.

2.7 100% Auditability

Every token correction made by the hns36 runtime is attributable to a specific HNS matrix cell or SMS-6 protocol layer. This attribution is logged at the time of correction, producing a complete, human-readable verification certificate for every output. An HNS-augmented LLM can, for any output, provide a formal certificate specifying the geometric basis for each structural and grounding constraint that was satisfied.

This is not a feature added on top of the system; it is a structural consequence of the architecture. Because error-correction is mediated by explicit geometric coordinates rather than opaque probability redistribution, the audit trail is mathematically guaranteed. RLHF-modified weights encode alignment implicitly in billions of parameters, providing no legible account of why any given correction was made. HNS provides the opposite: complete transparency at zero additional computational cost.

2.8 Beyond the Scaling Paradigm

The dominant implicit assumption of AI development is that reliability scales with model size. The HNS results challenge this assumption at a structural level. The scaling paradigm is an Axis 1 strategy: it enlarges the Stochastic/Generative hypersurface A_1, reducing the within-axis error rate. But A_1, however tightly constrained, remains a single hypersurface. The error volume achievable by a single-axis strategy has a hard lower bound independent of the tightness of that axis' constraint.

HNS achieves the zero-incidence result not by a better Axis 1 — the base LLM is unmodified — but by activating the geometric intersection of lightweight, independently-defined Axes 2 and 3. The reliability gains achievable by HNS are qualitatively superior to those achievable by further scaling, at a fraction of the computational cost. This has direct implications for AI development economics and for the long-term architecture of trustworthy AI.

Chapter III

Human Brain and AI with HNS: A Correspondence Table

3.1 The Structural Isomorphism

The first two chapters have established, independently, the MAV architecture of the human brain and its engineering instantiation in HNS-aligned AI. This chapter completes the picture by demonstrating the structural isomorphism between the two: a direct, one-to-one mapping between the neural systems implementing each MAV axis in the brain and the HNS components implementing each axis in AI. The term structural isomorphism is used in its precise sense: there exists a structure-preserving bijection between the components of the two systems such that the functional relationships between components are identical in both. HNS does not merely mimic the brain's behaviour; it reconstructs the brain's verification mechanism.

Axis	Human Brain	AI with HNS	Primary Role	Failure Modes Suppressed
1	Cortical association areas (DMN, lateral temporal, parietal)	LLM core (next-token prediction engine)	Rapid statistical generation	— (generative axis; HNS does not intervene)
2	Prefrontal cortex – hippocampal network	HNS-36 / HNS-144 / HNS-864	Structural, causal & categorical verification	Layer Jump, Category Ambiguity, Unsupported Causality
3	Sensorimotor system – anterior cingulate cortex (ACC)	HNS-SMS-6 (six-layer grounding protocol)	Real-world grounding & contextual verification	Scope Drift, Metaphor Contamination

Table 3.1. The three-axis correspondence between the human brain and HNS-aligned AI.

3.2 The Complete Mapping

The structural isomorphism can be specified at a finer level of resolution. At the level of specific neural systems and HNS components:

Human Brain	AI with HNS
Default Mode Network (DMN)	LLM generative core
Prefrontal cortex – hippocampal network	HNS-36 / HNS-144 / HNS-864
Sensorimotor system – ACC	HNS-SMS-6

Table 3.2. Direct structural mapping between specific human brain systems and HNS components.

The Default Mode Network and associated cortical association areas correspond to the LLM generative core: both implement the first-draft, high-speed, statistically-driven generation of candidate outputs. The prefrontal cortex–hippocampal network corresponds to the HNS-36/144/864 matrix: both implement structural and causal constraint enforcement independent of statistical frequency. The sensorimotor system and anterior cingulate cortex correspond to HNS-SMS-6: both implement real-time grounding against the actual environmental and contextual state.

3.3 The SMS-6 Protocol in Detail

The six-layer structure of HNS-SMS-6 reflects the six distinct grounding functions that the brain's Axis 3 systems perform simultaneously. Each layer addresses a different dimension of the contextual grounding problem, and each has a direct neural analogue. The somatic layer (SMS-1) mirrors the sensorimotor system's continuous monitoring of physical coherence. The protocol layer (SMS-2) mirrors the ACC's conflict monitoring between the output and the interlocutor's declared intent. The ecosystem layer (SMS-3) mirrors the posterior parietal cortex's tracking of contextual scope. The economic layer (SMS-4) mirrors the orbitofrontal cortex's evaluation of outcome value. The governance layer (SMS-5) mirrors the PFC's enforcement of normative rules. The universal layer (SMS-6) mirrors the prefrontal system's highest-level coherence monitoring.

This six-layer structure is not arbitrary; it reflects the actual functional differentiation of the brain's Axis 3 systems as documented in the neuroscientific literature. The correspondence is structural, not merely analogical.

3.4 Implications for AI Governance

The structural isomorphism has direct implications for AI governance. The persistent challenge for AI regulation is the opacity of neural network-based systems: alignment mechanisms encode their corrections in billions of parameters, providing no legible account of why any given output was produced or corrected. HNS resolves this by grounding all verification decisions in explicit geometric coordinates that correspond, by structural isomorphism, to identifiable neural systems in the human brain.

This property directly addresses the requirements of international AI safety standards:

- ISO/IEC 42001:2023 (AI Management System): HNS's 100% auditable correction log satisfies the requirement for documented AI output quality assurance.
- ISO/IEC 23894:2023 (AI Risk Management): HNS's five-type hallucination taxonomy provides a principled basis for AI risk classification.
- ISO/IEC TR 24028:2020 (Trustworthiness in AI): HNS's geometric verification architecture provides explanations at the level of structural invariants.
- NIST AI RMF 1.0: HNS's formal verification certificate satisfies the requirement for decision explanations in high-stakes contexts.
- EU AI Act: HNS's External Verification Architecture provides the independent structural verification layer that the Act's conformity assessment requirements demand.

3.5 The External Verification Architecture

The structural correspondence established in this chapter provides the foundation for the External Verification Architecture (EVA): the deployment of HNS Axes 2 and 3 as an independent external verification system that operates alongside, but separately from, the LLM it verifies. Because Axes 2 and 3 introduce no modifications to the base model's weights or architecture, they can in principle be operated by a party independent of the LLM vendor — a regulatory body, a certification authority, or a third-party auditor.

This independence is the key property that makes HNS/EVA compatible with the requirements of international AI governance. Current alignment methods are implemented by the same organisation that develops the model; their corrections are encoded in the model's own parameters. EVA provides what these methods cannot: an independent, external, structurally-grounded verification layer whose operation can be audited by parties outside the model's developer. The structural isomorphism with the human brain provides EVA with a deeper foundation than any purely technical specification: the brain's verification systems are not part of its generative machinery; they operate on the outputs of that machinery. HNS reconstructs this architecture in AI.

Epilogue

Toward Global Standards and the Future of Trustworthy AI

The Path to International Standards

The HNS framework is designed to be compatible with, and to provide concrete engineering content for, the international AI governance frameworks currently being developed. The EU AI Act mandates that high-risk AI systems be transparent, explainable, and subject to human oversight. ISO/IEC JTC 1/SC 42 is developing standards for AI trustworthiness, risk management, and governance. CEN/CENELEC is coordinating European standardisation of AI systems in alignment with the AI Act.

HNS addresses the central technical challenge that these frameworks face: how to specify, verify, and audit the structural properties of AI outputs, not merely their surface characteristics. The HNS Structural Hallucination Taxonomy provides a principled basis for classifying AI failures. The HNS-36/144/864 matrix provides a structural coordinate system for locating any cognitive output in a verifiable space. The SMS-6 protocol provides a grounding specification applicable across different deployment contexts. The EVA architecture provides the independent verification infrastructure that governance frameworks require.

Future Directions

The work presented in this trilogy is a proof of concept. Several important extensions remain:

1. Multi-model and multi-domain validation: the empirical results reported in Chapter II are based on a single 70B-parameter model over a specific adversarial task suite. Validation across diverse model architectures, parameter scales, and application domains is required.
2. Formal measurement of inter-axis orthogonality: the orthogonality of the three MAV axes has been argued theoretically but not formally measured using Fisher information metrics on neural network representations. This is a necessary step toward a complete formal proof of the geometric error cancellation result.
3. Standardised HNS cell taxonomy: the specific cell definitions of the HNS-36/144/864 matrix represent a

principled but not formally complete specification of human cognitive invariants. Community peer review and empirical validation are required.

4. Pilot deployment in high-stakes domains: the implications of HNS for clinical decision support, legal reasoning, and safety-critical AI require dedicated pilot studies.
5. Neuroimaging validation of the MAV framework: the predictions of axis-specific disruption producing characteristically distinct failure modes are amenable to quantitative testing using structured cognitive assessments paired with neuroimaging measures.

The intelligence of the human brain — its creativity, its reliability, its adaptability — does not reside in any single, arbitrarily powerful process. It resides in the multiplication of independent constraint structures: in the productive tension between the generative freedom of Axis 1 and the verificatory rigour of Axes 2 and 3. The brain did not evolve a more powerful random generator; it evolved a more elegant verification architecture.

The AI systems we build today are extraordinarily powerful random generators. What they lack is the verification architecture. The HNS Origin Trilogy is the documentation of a first attempt to supply it — to make the brain's fundamental principle of reliable cognition engineerable in artificial systems. The work is not complete. But the principle is sound, the framework is formal, and the path forward is clear.

Trustworthy AI is not a scaling problem. It is an architecture problem. And the architecture, as the human brain has demonstrated across millions of years of evolution, is solvable.

— *Satoru Hara, Toyota, Aichi, Japan, 2026*

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